

# **ANGLE LAKE 2020**

## **Lake Stewardship Monitoring Report**

King County Water & Land Resources Division  
Science & Technical Support Section  
[www.kingcounty.gov/EnvironmentalScience](http://www.kingcounty.gov/EnvironmentalScience)

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## **Summary & Recommendations**

*Thank you to Diane & Buff Chace, and Bruce & Sandy Goulet, the volunteer monitors for Angle Lake.*

**The key takeaways from the 2020 monitoring season are:**

- Angle Lake continued to have clear water, with low nutrient concentrations and low algal growth.
- Angle Lake continues to be one of the clearest lakes in the Lake Stewardship Program.
- No algal blooms were reported for toxin testing in 2020.

**The Lake Stewardship Program recommends:**

- Monitoring is a key part of good lake stewardship, building a valuable long-term dataset to guide lake management and detect any future problems. Continue to monitor Angle Lake through the Lake Stewardship Program.
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## **What We Measure & Why**

- **Secchi depth** is a measure of water clarity or transparency. Secchi depth is shallower when there are more suspended particles in the lake, such as sediment or algae. Secchi depth is also affected by water color, often from tannins or other naturally occurring organic molecules.
- **Water temperature** can affect the growth rates of plants and algae. In addition, cooler or warmer water temperatures favor different species of fish and other aquatic organisms. Many lakes in King County naturally have cold water, so increases in water temperature can favor non-native species.
- **Chlorophyll-a** is a measure of the amount of algae in a lake. Chlorophyll-a is a pigment necessary for algae to photosynthesize and store energy.
- **Phosphorus** and **nitrogen** are naturally occurring nutrients necessary for growth and reproduction in both plants and animals. Increases in nutrients (especially phosphorus) can lead to more frequent and dense algal blooms.
- The **ratio of total nitrogen to total phosphorus (N:P)** indicates whether nutrient conditions favor the growth of cyanobacteria (blue-green algae). When N:P ratios are near or below 25, cyanobacteria can dominate the algal community. This is important because cyanobacteria have the ability to produce toxins.

## **Water Quality Results & Trends**

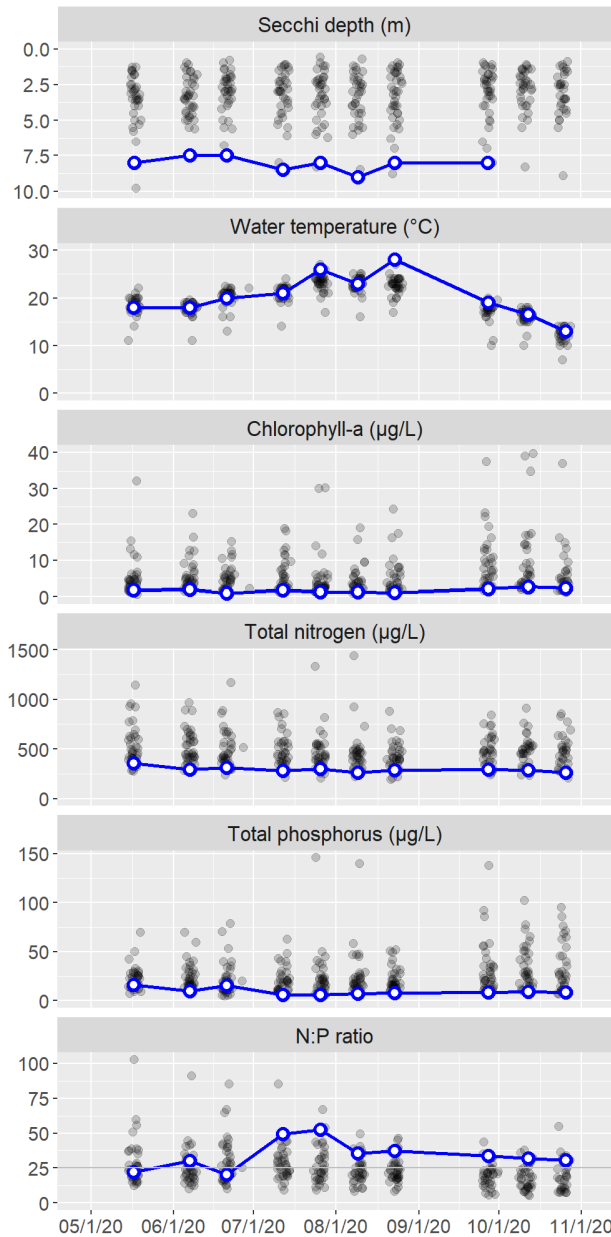
The following graphs show the water-quality parameters that are sampled from May through October, at 1 m depth (additional depths and parameters are measured on profile days; see *Supplemental Data*). The left column of graphs shows results for each sampling date in 2020, and the right column shows average values for each year (May-October averages).

Data for Angle Lake are the blue circles (with white centers) connected by the blue line. Any gaps in the blue line indicate missed samples. To provide some context for these values, the grey points in the background are results for all other lakes in the Lake Stewardship program.

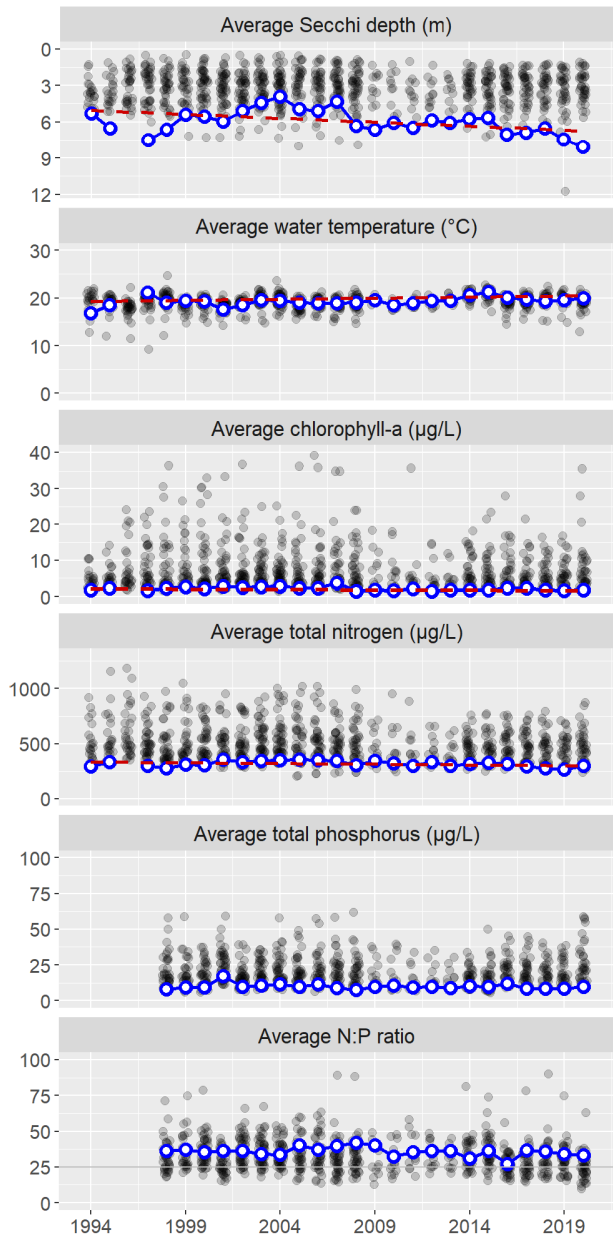
Any long-term trends in Angle Lake are drawn with a dashed red line and described further after the graphs. Statistical trend analyses used a seasonal (monthly) Kendall test ( $p < 0.05$ ).

In 2020, samples were collected at 10 out of the usual 12 sampling events throughout the summer. The first May event was missed as we worked out appropriate safety precautions and lab coordination. And the first September event was canceled due to the heavy smoke around the region, making it unsafe for our volunteers to be outdoors.

## 2020 Monitoring Results



## Long-Term Annual Averages



Nutrient and chlorophyll concentrations remained low and fairly stable throughout the monitoring season.

The table below gives more details about the long-term trends. Results are presented as an average amount and percent of change per decade (the increase or decrease over ten years). Percent change is calculated as the percent of the estimated value in 1994, when monitoring started.

| Parameter         | Change per Decade | (%)     |
|-------------------|-------------------|---------|
| Secchi depth      | 0.66 m            | (13%)   |
| Water temperature | 0.48 °C           | (2.5%)  |
| Chlorophyll-a     | -0.21 µg/L        | (-10%)  |
| Total nitrogen    | -12 µg/L          | (-3.8%) |

Chlorophyll concentrations have decreased over time in Angle Lake, and Secchi depths have gotten deeper, suggesting that water quality has been improving slightly. Although the 0.21 µg/L decrease per decade would be insignificant in many lakes, it is several percent of Angle Lake's average chlorophyll concentration.

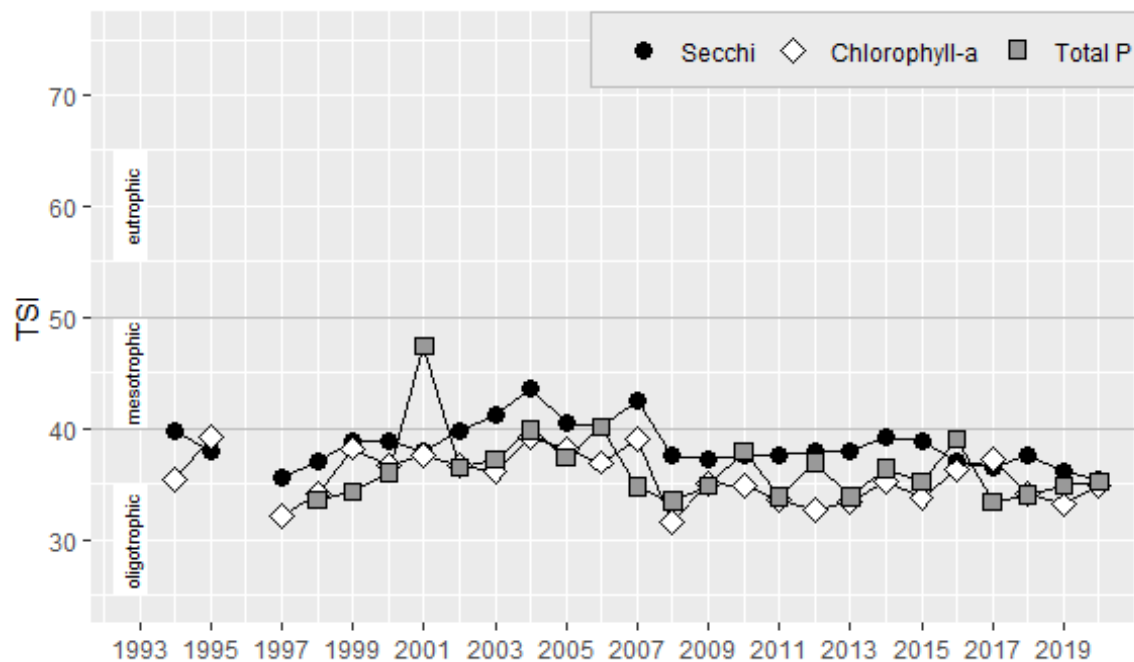
## **Trophic State**

The Trophic State Index (TSI) is a common index of a lake's overall biological productivity. TSI values are calculated from Secchi depth, chlorophyll-a concentrations, and total phosphorus concentrations. These three TSI estimates are all scaled between 0 and 100.

TSI calculations use average values from June-September, focusing on fairly consistent "summer" conditions. This is in contrast with the annual averages shown above, which also include May and October data.

*Oligotrophic* lakes (TSI <40) are very clear, with low nutrient concentrations and low algal growth. *Eutrophic* lakes (TSI >50) have less-clear water, with high nutrient concentrations and high algal growth. Eutrophic lakes are more likely to have frequent algal blooms. *Mesotrophic* lakes (TSI 40-50) are in the middle, with fairly clear water, and moderate nutrient concentrations and algal growth. Lakes in lowland King County have a range of different natural trophic states, and human activities may also alter a lake's trophic state (usually by changing nutrient inputs).

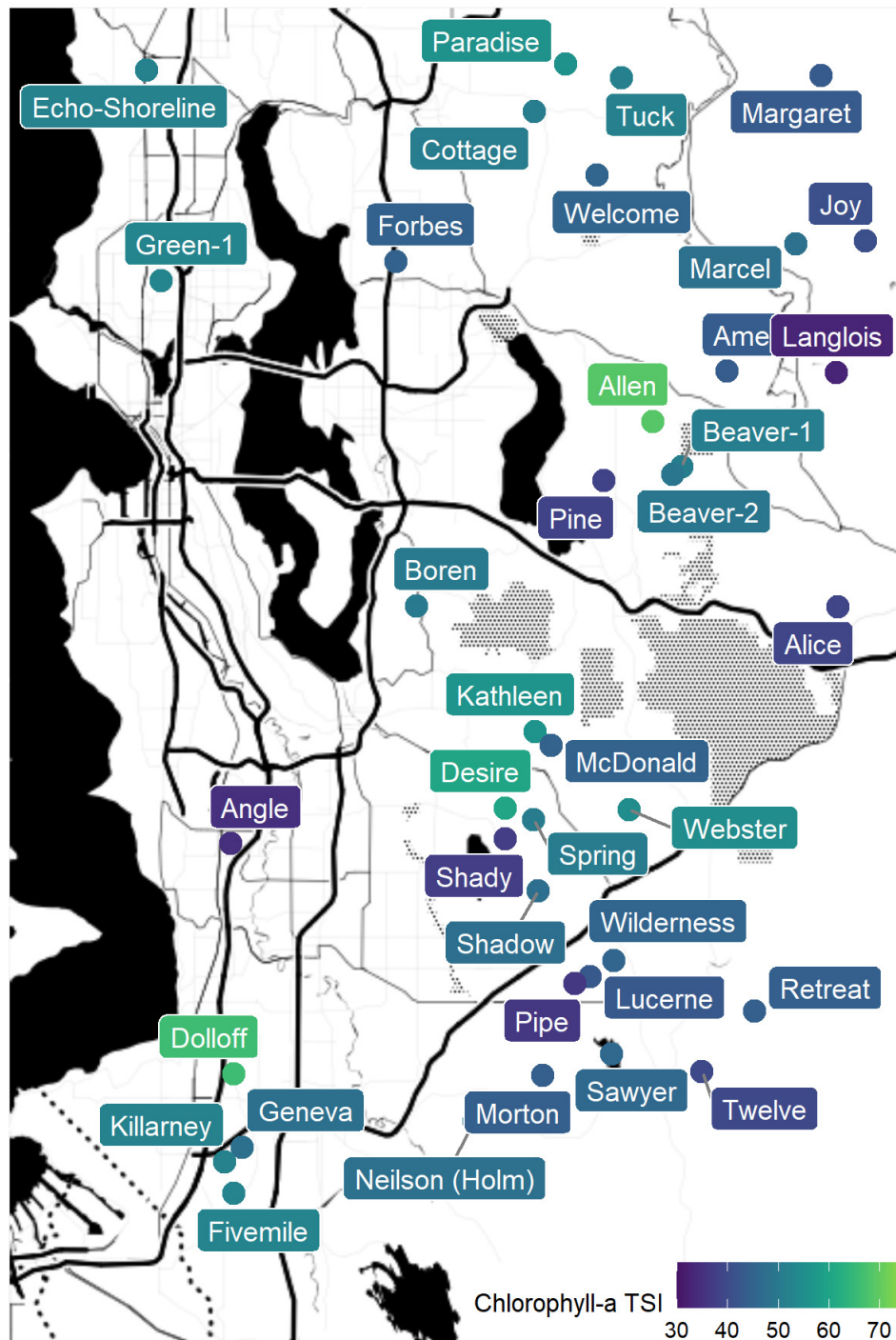
## **Trophic state indices**



In 2020, all three TSI values were in the oligotrophic range.

## Comparison map

For a comparison with other lakes, this map shows the trophic state for each lake in the King County Lake Stewardship program in 2020. The color of each circle indicates the lake's average chlorophyll-a TSI value for the year.



## **Supplemental Data**

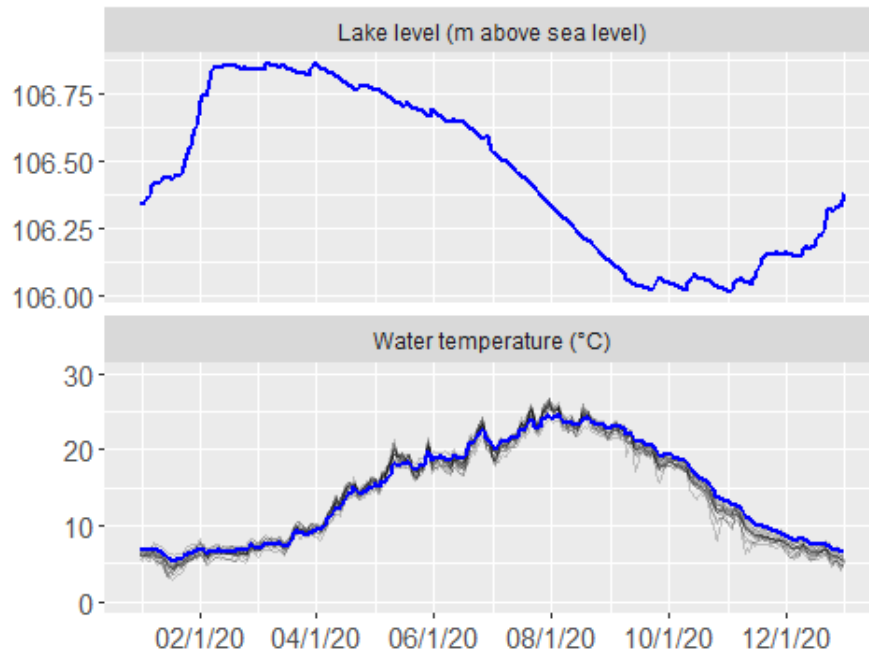
### **Summary statistics**

This table summarizes data from 2020 (1 m depth only), giving the minimum, mean (average), and maximum values for each parameter. This includes summary statistics for the full 2020 calendar year for Secchi and temperature, which were measured year-round, and May-October summary statistics for all parameters. To reduce biases from missing data or changes in sampling frequency, monthly means were calculated and then averaged to give an overall mean.

| Parameter                     | Minimum | Mean  | Maximum |
|-------------------------------|---------|-------|---------|
| <b>Full-year statistics</b>   |         |       |         |
| Secchi depth (m)              | 4.0     | 6.3   | 10.1    |
| Water temperature (°C)        | 5.0     | 16.4  | 28.0    |
| <b>May-October statistics</b> |         |       |         |
| Secchi depth (m)              | 5.8     | 7.4   | 10.1    |
| Water temperature (°C)        | 13.0    | 20.0  | 28.0    |
| Chlorophyll-a (µg/L)          | 1.0     | 1.8   | 2.6     |
| Total nitrogen (µg/L)         | 261.0   | 298.9 | 354.0   |
| Total phosphorus (µg/L)       | 5.7     | 9.9   | 16.0    |
| N:P ratio                     | 20.2    | 33.2  | 52.6    |

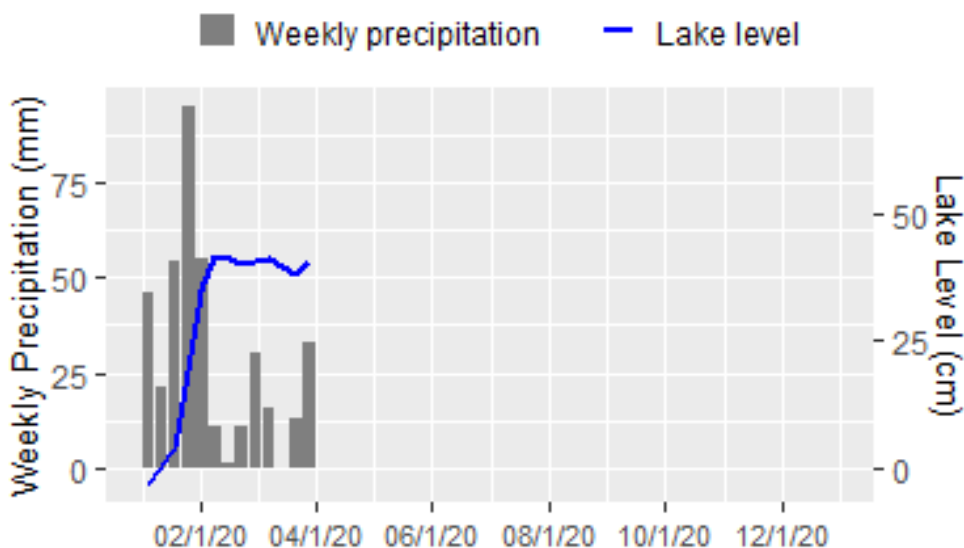
## Continuous lake level and temperature

Lake level and water temperature were recorded continuously by an automated logger. The blue line shows daily averages for Angle Lake. Grey lines in the background are temperatures for all other lakes with loggers.



## Hydrology: Lake level and precipitation

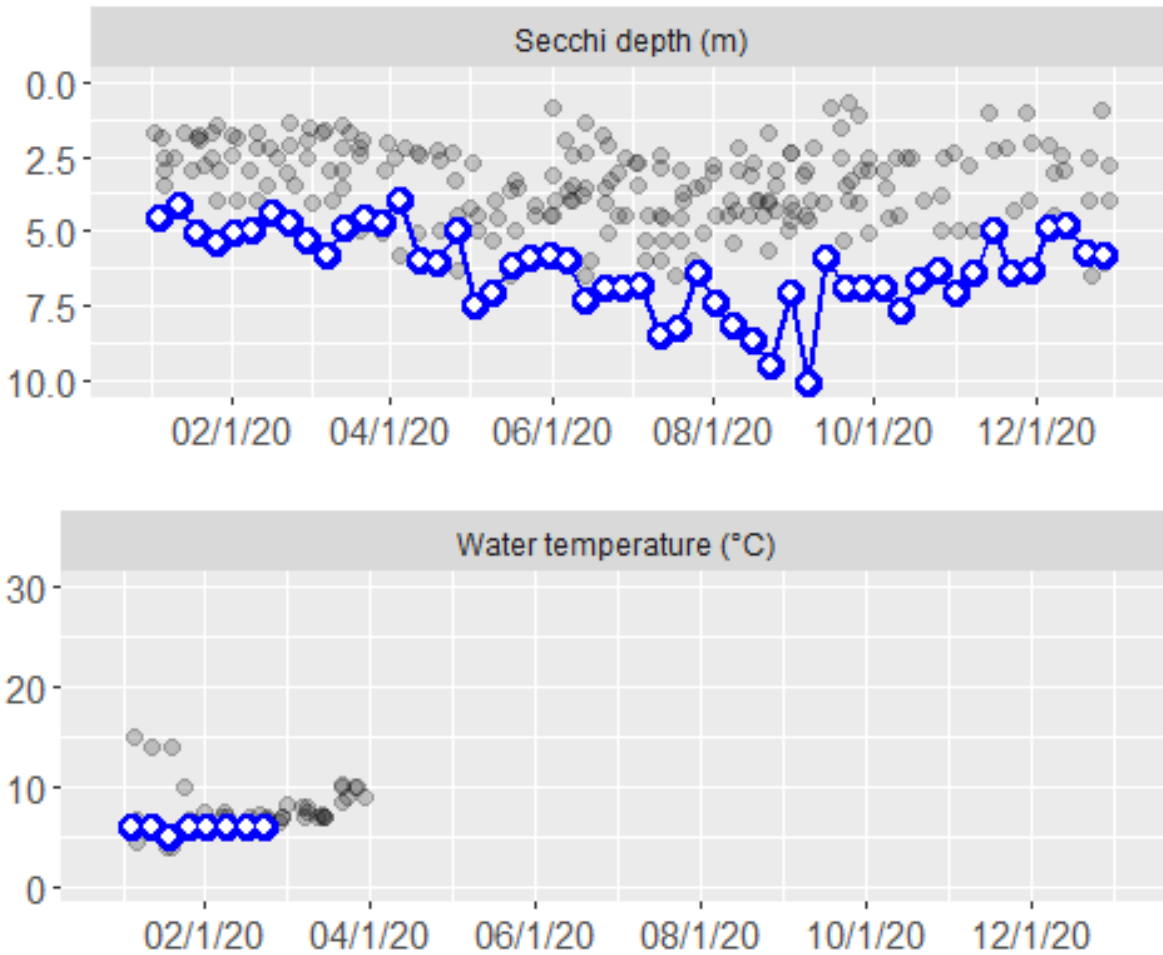
Lake level and precipitation were measured year-round. Bars show total weekly precipitation, and the line shows average weekly lake level.





## Year-round Secchi depth and water temperature

Secchi depth and water temperature (at 1 m depth) were measured weekly in 2020. The blue circles (with white centers) and blue line are data for Angle Lake. Gaps in the line indicate missed sampling dates. Grey points in the background are results for all other lakes in the Lake Stewardship program.



## Water column profile

In June and August, water was collected at the mid-lake sampling station from three depths in a water-column profile: 1 m, the middle depth of the water column, and 1 m from the lake bottom.

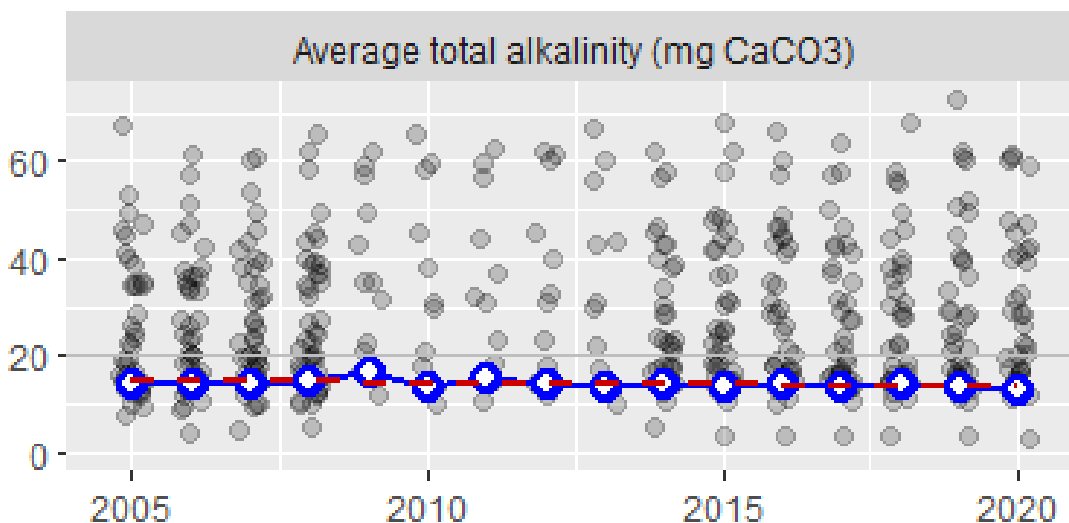
| Date      | Depth | Temp | Chlor | Pheo  | TN   | NH3   | NO2/3  | TP    | OPO4  |
|-----------|-------|------|-------|-------|------|-------|--------|-------|-------|
| 6/21/2020 | 1     | 20.0 | 1.0   | (1.0) | 311  | 6.1   | (10.0) | 15.4  | (0.5) |
|           | 7     | 18.0 | 3.3   | (1.0) | 323  | –     | –      | 14.2  | –     |
|           | 14    | 9.0  | –     | –     | 836  | 432.0 | (10.0) | 71.0  | 11.0  |
| 8/23/2020 | 1     | 28.0 | 1.1   | (1.0) | 288  | 3.6   | (10.0) | 7.7   | 0.5   |
|           | 7     | 22.5 | 1.8   | (1.0) | 354  | –     | –      | 17.8  | –     |
|           | 14    | 9.0  | –     | –     | 1670 | 344.0 | (10.0) | 207.0 | 9.8   |

Parameter abbreviations are: chlorophyll-a (Chlor), pheophytin (Pheo), total nitrogen (TN), ammonia (NH3), nitrate/nitrite (NO2/3), total phosphorus (TP), orthophosphate (OPO4). Depth is in m, temperature is in °C, and all other parameters are in µg/L. Dashes indicate parameters that were not analyzed for a given sample. Values below the method detection limit (MDL) are enclosed in parentheses and have the value of the MDL substituted.

## Total alkalinity

A lake's ability to resist acidification, also called its buffering capacity, is measured as "total alkalinity." Lakes with total alkalinity less than 20 mg CaCO<sub>3</sub> are considered sensitive to acidification. We measured total alkalinity in May and August (on profile-sampling days) at 1 m depth. In 2020, the average total alkalinity of these two samples was 13.3 mg CaCO<sub>3</sub>.

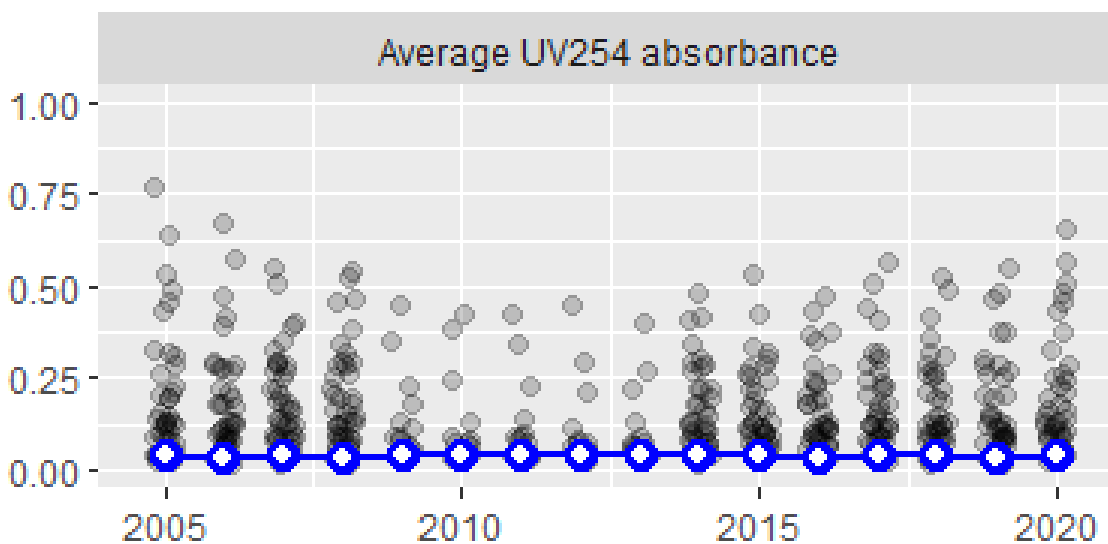
The blue circles (with white centers) and blue line are annual average alkalinity values for Angle Lake. Grey points in the background are results for all other lakes in the Lake Stewardship program. The dashed red line shows the long-term trend in alkalinity, with an average change of -1 mg CaCO<sub>3</sub> (-6.6%) per decade.



## Water color

Water color affects a lake's water clarity (and Secchi depth). Water color is measured by shining a specific wavelength of ultraviolet light (254 nm) through a filtered water sample and measuring the percent that was absorbed. We measured UV254 absorbance in May and August (on profile-sampling days) at 1 m depth. In 2020, the average UV254 absorbance of these two samples was 0.04, on a scale where 0 is no absorbance (perfectly clear) and 1 is complete absorbance (perfectly opaque).

The blue circles (with white centers) and blue line are annual average UV absorbance values for Angle Lake. Grey points in the background are results for all other lakes in the Lake Stewardship program.



Visit the [King County Lake Stewardship](#) website for more data and information. Data from automated loggers is on the [King County Hydrologic Information Center](#) website.



**King County**

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